Nagar Yuwak Shikshan Sannstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur.

Department of Information Technology

Odd Term : 2025-26
Semester : V
Subject:Operating System
Time: 1 ½ Hrs.

Mid Semester Examination-I
Date: 15/09/2025
Subject Code:- 23IT1506
Max. Marks: 30

Instructions:- 1. Give appropriate diagrams wherever necessary
2. Zero marks will be allotted if question number is not mentioned correctly.

Q 1		**SOLVE ANY TWO:**		
	a)	Describe the layered structure of an operating system. How does this architecture improve modularity and maintainability?	[5]	CO1(L2)
	b)	Differentiate between user mode and kernel mode. Why is dual-mode operation essential for system security?	[5]	CO1(L4)
	c)	Define system calls. How do they act as an interface between user programs and the operating system?	[5]	CO1(L1)
Q 2		**SOLVE (a) and (b) or (c):**		
	a)	Explain the structure and role of the Process Control Block (PCB) in process management.	[5]	CO3(L2)
	b)	What is a process context switch? Explain its steps and impact on system performance.	[5]	CO3(L2)
	c)	Discuss the advantages & disadvantages of FCFS, SJF, Nonpreemptive priority and RR and solve the following.Consider the following set or processes, with the length of the CPU burst given in milliseconds:	[10]	CO2(L3)

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

a. Draw four Gantt Charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, nonpreemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 4).
b. What is the turnaround time of each process for each of the scheduling algorithms in part a?
c. What is the waiting time of each process for each of these scheduling algorithms?
d. Which of the algorithms results in the minimum average waiting time (over all processes)?

Q 3		**SOLVE ANY TWO:**		
	a)	Define race condition. How does it affect the correctness of a concurrent system?	[5]	CO3(L1)
	b)	Explain the concept of mutual exclusion. How does it help in solving synchronization problems?	[5]	CO3(L2)
	c)	Explain the four necessary conditions for a deadlock to occur. Can a deadlock happen if even one of these conditions is absent? Justify your answer.	[5]	CO3(L2)

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Department of Information Technology

Odd Term : 2025-26 | **Mid Semester Examination-II**

Semester : V | **Date: 15/11/2025**

Subject:Operating System | **Subject Code:- 23IT1506**

Time: 1 ½ Hrs. | **Max. Marks: 30**

Instructions:- 1. Give appropriate diagrams wherever necessary
2. Zero marks will be allotted if question number is not mentioned correctly.

Q 1		**SOLVE ANY TWO:**		
	a)	Compare contiguous, linked, and indexed allocation methods. Which is most efficient for random access and why?	[5]	CO3(L4)
	b)	The queue of pending requests, in FIFO order, is: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms? a. FCFS b. SSTF c. SCAN d. LOOK e. C-SCAN D – 4999 Head = 143	[5]	CO3 (L3)
	c)	Describe the first fit, best fit, and worst fit strategies for disk space allocation and solve the following: Given five memory partitions of 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of 212 KB, 417 KB, 112 KB, and 426 KB (in order)? Which algorithm makes the most efficient use of memory?	[5]	CO3 (L2)
Q 2		**SOLVE ANY TWO:**		
	a)	Distinguish between static and dynamic memory partitioning? Illustrate with diagrams.	[5]	CO4(L4)
	b)	Desribe Translation Lookaside Buffer (TLB)? Why is it used in paging systems?	[5]	CO4(L2)
	c)	Describe the structure and function of a page table. How is it used in address translation?	[5]	CO4(L2)
Q 3		**SOLVE ANY TWO:**		
	a)	Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6 How many page faults would occur for the following replacement algorithms, assuming one, two, three, four, five, six, and seven frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each. • LRU replacement • FIFO replacement • Optimal replacement	[5]	CO4 (L3)
	b)	Describe the steps involved in handling a page fault in a demand paging system.	[5]	CO4(L2)
	c)	Compare tree-structured and acyclic graph directory systems.	[5]	CO4(L4)

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Mid Semester Exam - I

ODD Term—2025-26
Semester – V
Subject Code: 23IT502
Time : 1½ Hours

Date: - 10/09/2025
Sub: DAIS
Max. Marks: 30

Instructions to examinees:
Attempt all questions.
Figures in bracket to the right indicate the marks for questions.

Q.1	**Solve any Two**	**10**	**CO mapped**
a)	Compare ER Model and EER Model with suitable examples.	05	CO1,L2
b)	Give the concept of Data Abstraction ?Also,Explain the different levels of abstraction with diagram in a database system.	05	CO1,L2
c)	A Travel agency maintain various taxies the list of drivers and their licence no.also maintain by them.A logbook maintains details about tools that might have taken place along with data location and drivers name and other details . 1.Construct ER MODEL 2.Convert ER into relational model and Identity key attributes. 3.Construct table with this ER Model.	05	CO1, L3
Q.2	**Solve any Two**	**10**	
a)	What are aggregate functions? Enlist the aggregate functions supported by SQL giving appropriate example.Also, difference between WHERE and HAVING Clause.	05	CO2,L2
b)	Consider the following relational database: employee(emp_no ,ename,salary,dept_no,job_title) department(dept_no ,dname,location) For each of the query given below, give an expression in SQL. a)List the employee name and department name for employee whose salary is greater than1500. b)For each department number find how many employees are working. c)Find employees who earn more than 60,000. d) Find the maximum salary in each department e) List all employees and their department names	05	CO2,L2
c)	Elaborate the concept of Join? Explain Inner and Outer types of Joins with an example.	05	CO2, L3
Q.3	**Solve any Two**	**10**	
a)	• Apply normalization to the following relation: `STUDENT (Stud_ID, Stud_Name, Course, Instructor, Instructor_Phone)` • Analyze and normalize it up to 3NF.	05	CO3,L3
b)		05	CO3,L2
b)	Explain all types of Normal forms, Differentiate between 3NF and BCNF.		
c)	Explain in detail the concept of functional dependency and state significance of normalization.	05	CO3,L3

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Department of Information Technology

ODD Term—2025-26
Semester – V
Subject Code : 23IT1501

Mid Semester Exam II
Date: - 13/11/2025
Subject Theory of Computation

Time :1½ Hours

Max. Marks: 30

Instructions:

1. All questions are compulsory
2. Give appropriate figure where ever necessary
3. Marks will be allotted as zero if wrong question number is mentioned.

Q.1.(Any Two)	a) Design Push Down Automata for the language $L = \{ a^n b^{2n} \mid n>=1$ by empty stack}	[05]	CO4 (Level 6)
	b) Construct Push Down Automata for the following context free grammar S→ 0BB B→ 0S | 1S | 0 Test whether "010000" is in N (A)	[05]	CO4 (Level 6)
	c) Design Push Down Automata for the language L L = {x | x consists of equal number of a's and b's}.	[05]	CO4 (Level 6)
Q.2.(Any Two)	a) Construct Turing Machine that can accept all strings over {a, b} containing odd Number of a's.	[05]	CO4 (Level 6)
	b) Design Turing Machine to find 2's Complement of given binary number.	[05]	CO4 (Level 6)
	c) Show that the union of two recursively enumerable languages is recursively enumerable	[05]	CO4 (Level 3)
Q.3. (Any Two)	a) Write Ackerman function. Apply this function to compute A (1, 3) and A (2, 1).	[05]	CO3 (Level 3)
	b) When a language is said to be recursive? Is it true that every regular set is not recursive	[05]	CO3 (Level 3)
	c) State Post Correspondence problem. Find whether the lists A=(11,100,111) and B= (111, 001, 11) have a Post Correspondence Solution ?Write its applications.	[05]	CO3 (Level 3)

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Department of Information Technology

ODD Term—2025-26
Semester – V
Subject Code : 23IT1501

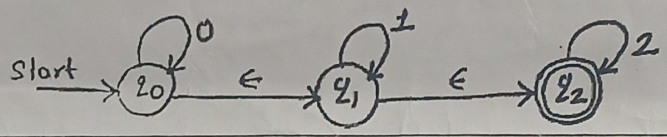
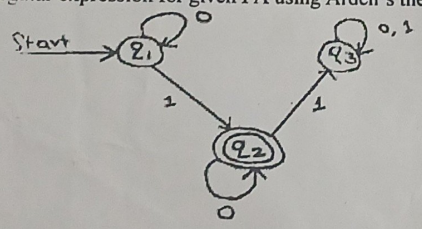
Mid Semester Exam I
Date: - 12/09/2025
Subject Theory of Computation

Time :1½ Hours

Max. Marks: 30

Instructions:

1. All questions are compulsory
2. Give appropriate figure where ever necessary
3. Marks will be allotted as zero if wrong question number is mentioned.

Q.1.(Any Two)	a) Design a DFA ending with 101, $\Sigma=\{0,1\}$	**[05]**	CO2 (Level 4)
	b) Construct DFA equivalent to the following M= ({q0, q1, q2, q3}, {a,b}, δ, q0,{ q3}) where δ is given as δ: q0 → {q0, q1}, q0 / q1 → q2, q1 / q2 → q3, q3 / *q3 → -, q2	**[05]**	CO2 (Level 2)
	c) Convert the following NFA with ε to NFA without ε moves 	**[05]**	CO2 (Level 2)
Q.2.(Any Two)	a) Write algortihm to convert left linear grammar to right linear grammar. Apply the algortihm on given grammaer to convert it to right linear grammar. S→Aa/ a/ b A→Bb/ Ca / a C→c	**[05]**	CO2 (Level 2)
	b) Find regular expression for given FA using Arden's theorem. 	**[05]**	CO2 (Level 2)
	c) List and explain types of grammars in the Chomsky Hierarchy?	**[05]**	CO2 (Level 1)
Q.3. (Any Two)	a) Consider the grammar given as, G=({S, A}, {a, b}, P, S) where 'P' consisting of , S→AaS/a A→SbA/SS/ba Derive "aabbaa" using left most and rightmost derivation. Also draw left most and rightmost derivation tree.	**[05]**	CO2 (Level 2)
	b) What is ambiguous grammar? Check whether the given grammar is ambiguous or not- S → SS S → a S → b Consider w = abba	**[05]**	CO2 (Level 3)
	c) How to remove useless symbol from the grammar? Test whether the grammar contains useless symbol, if yes eliminate it. S→aS/ A/ C A→a B→aa C→aCa	**[05]**	CO2 (Level 3)

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ODD Term—2025-26
Semester – V
Subject Code: 23IT1502
Time : 1½ Hours

Mid Semester Exam - II
Date: - 11/11/202 5
Subject: Database & Information System
Max. Marks: 30

Instructions:

1. Give appropriate figure where ever necessary
2. Marks will be allotted as zero if wrong question number is mentioned.

Q.1.	**Solve any Two**		
	a) What is meant by the term Query Processing? & discuss various steps involved in this process?	[05]	CO2(L2)
	b) Elaborate three techniques commonly used when developing algorithms for relational operators. Explain how these techniques can be used to design algorithms for the selection, projection, and join operators.	[05]	CO2(L2)
	c) Demonstrate the External Sort-Merge Algorithm with diagram. and associated example to show it.	[05]	CO2(L2)
Q.2.	**Solve any Two**		
	a) Explain how each ACID property is maintained internally by a database management system when a system crash occurs during a transaction.	[05]	CO3(L1)
	b) Justify whether the given schedule S is conflict serializable or not. If yes, then determine all the possible serialized schedules. T1: W(B); T2: R(A), W(A), W(B); T3: R(A), R(B); T4: R(A)	[05]	CO3(L3)
	c) Elaborate different between 1. View and Conflict schedules. 2. Recoverable and Non – Recoverable schedules.	[05]	CO3(L2)
Q.3.	**Solve any Two**		
	a) Describe the strict 2 PL two phase locking protocol in brief.	[05]	CO4(L4)
	b) What is deadlock? Explain different deadlock prevention techniques	[05]	CO4(L4)
	c) Explain Triggers and its types with significance & Write an SQL query to create a trigger that automatically updates the updated_at column of a table named employees whenever a record is updated.	[05]	CO4(L1)

Department of Information Technology
EVEN Term (Session 2025-26)
Mid Semester Examination-II
Subject: Machine Learning [23IT1504]

Date:14.11.25
Max. Marks: 30

Semester: VI
Time: 1.5 Hrs.

Solve any two sub questions from each question

Qu 1	a)	What is semi supervised learning? Suppose an e-commerce company has: ◊ 1000 customer reviews about a product ◊ Only 100 reviews are manually labelled as positive (1) or negative (0) ◊ 900 reviews are unlabelled (no one had time to tag them) We want to train a model that classifies *new reviews* as positive or negative, using all the 1000 examples. Explain stepwise procedure using semi-supervised learning to achieve this.	05	CO2	L2
	b)	What is divisive clustering? Perform Divisive clustering using Minimum Spanning Tree for the following proximity matrix, where A, B, C, D are data points:	05	CO2	L2
	c)	What is agglomerative clustering? Perform agglomerative clustering on the following dataset and report the data points in each cluster for k=2. Use Euclidean distance and Centroid Linkage. Dataset = [[1,3], [1,4], [5,2], [5,1], [2,2], [7,2]]	05	CO2	L2
Qu 2	a)	What is k-fold cross validation and why is it required? Suppose we have a dataset of 12 training samples, labeled X_1 to X_{12}. Apply 3-fold cross validation. How many models are trained and tested? How is the best model found using this method?	05	CO4	L5
	b)	Calculate MAPE and R^2 values for the Regression Model MR_1 that predicts the resale value of a car in lakh (Rs). Use the following true and target values of the test data: Y_true = [5, 6.5, 8, 4.4, 7.1] Y_pred = [4.8, 6.3, 7.5, 4.2,6.8] If another ML model MR_2 has Y_pred = [4.9, 6.35, 7.2, 4.05, 6.85], find which model is performing better on the test data?	05	CO4	L5
	c)	What is a confusion matrix?	05	CO4	L5
Qu 3	a)	Describe construction of MP Neuron. Implement 3-input OR gate using MP Neuron.	5	CO2	L3
	b)	What is the use of pooling layer in CNN? Apply Max pooling on the given 4 by 4 image with dropout mask of size 2 by 2 given below and stride equal to 2.	5	CO2	L3
	c)	What is a feedforward neural network? Find the value of O in the following ANN, if input vector is [-1,1] and MP neurons are used in the network.	5	CO2	L3

Confusion Matrix for the Multi-Class Machine Learning Model is given to the left. Evaluate the model by finding the accuracies of each of the three classes using one vs rest approach.

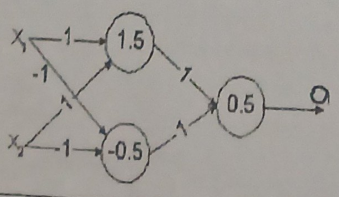
Actual Values \ Predicted Values	Setosa	Versicolor	Virginica
Setosa	16	0	0
Versicolor	0	17	1
Virginica	0	0	11

3	2	6	5
2	1	6	9
7	9	4	6
7	8	2	5

Image

1	0
0	1

dropout mask



Yeshwantrao Chavan College of Engineering, Nagpur
Department of Information Technology
ODD Term (Session 2025-26)
Mid Semester Examination-I
Subject: Machine Learning [23IT1504]

Semester: VI **Date:13.09.25**
Time: 1.5 Hrs. **Max. Marks: 30**

Solve any two sub questions from each question

Qu 1	a)	Differentiate between classification and regression. Consider a Credit Scoring application. The objective of credit scoring is to assign both existing and prospective customers to one of two groups: good or bad, depending on the applicant's demographics and credit bureau records and their recent financial transactions. A member of the good group is considered likely to repay their financial obligation like a loan. Is this a classification or regression problem? justify.	05	CO1	L2
	b)	What are the three types of Machine Learning algorithms? A publishing company wants to automatically organize a large collection of research articles into meaningful topics to help researchers easily browse articles related to their field of interest. They do not have any labeled data indicating the topic of each article. Which type of Machine Learning approach (Supervised or Unsupervised) is most suitable for this task and why? Give one commonly used algorithm that can be applied to this problem.	05	CO1	L2
	c)	A Multivariate linear model for Student's salary prediction for YCCE is trained and has the model coefficients [β_1 , β_2 , β_3] = [0.85, 0.86, 1.07] for $[CGPA, CommSkill, CodingSkill]$ respectively and $\beta_0 = -13.66$. Then answer the following: • Is it a regression or classification Model? • Write the equation of the model. • If a student has $x = [8.1, 8.0, 5.0]$, then what is his predicted salary ? • Can we use this model to predict the salary of a student of other colleges? Justify.	05	CO1	L2
Qu 2	a)	Consider the following table – it consists of the height, age and weight (target) value for 10 people. As you can see, the weight value of ID 11 is missing. Predict the weight of this person based on their height and age, using KNN with k=2 and the Manhattan Distance.	05	CO2	L3
	b)	Consider the dataset of some course students given in the following table. Use KNN algorithm to predict Final score for the student with study hours=11 and attendance=78%. Use the value of k = 3 and the Euclidean distance.	05	CO2	L3
	c)	What is feature independence assumption in Naïve Bayes classification? In a binary classification using Naive Bayes, a test instance has the following probabilities: P(Class A) = 0.4, P(Class B) = 0.6 P(x1|A) = 0.5, P(x2|A) = 0.2 P(x1|B) = 0.1, P(x2|B) = 0.8 Assuming feature independence, which class will the model assign to the instance with feature vector **x** = [x1, x2] ?	05	CO2	L3

ID	Height	Age	Weight
1	5	45	77
2	5.11	26	47
3	5.6	30	55
4	5.9	34	59
5	4.8	40	72
6	5.8	36	60
7	5.3	19	40
8	5.8	28	60
9	5.5	23	45
10	5.6	32	58
11	5.5	38	?

ID	Study Hours	Attendance	Final score
1	12	85	88
2	10	80	75
3	15	90	95
4	8	70	65
5	9	75	70
6	14	88	92
7	7	60	50
8	13	82	85

Qu 3	a)	A decision tree regression model is trained to predict a person's annual medical expenses in INR based on two predictors: Age (years) and BMI (Body Mass Index). The dataset used for training is :	5	CO3	L4

Age	BMI	Medical Expense
25	22.0	25000
35	28.0	40000
45	31.0	60000
50	29.0	55000
60	35.0	80000
30	25.0	30000
40	30.0	50000

- First split on **BMI** at 28.5:
  - If BMI $\leq$ 28.5 → Leaf Node A
  - If BMI > 28.5 → Next split on Age
- Next split on **Age** at 45:
  - If Age $\leq$ 45 → Leaf Node B
  - If Age > 45 → Leaf Node C

i) Draw the decision tree using these splits.
ii) Analyze the tree and find the predicted medical expense using mean for a patient aged **42 years** with **BMI = 29.0**

	b)	Consider the decision tree given below, that decides whether to play tennis or not in a given weather condition represented by feature vector [Temperature, Outlook, Wind, Humidity] and answer the following questions. It is given that there are 9 instances belonging to class YES (Play Tennis) and 5 instances belonging to the class NO (Do not play Tennis). Analyse the given tree and answer the following: i) How the node labelling (class labelling) takes place? ii) Redraw the tree with class labels written in place of "?" for each node. iii) Encircle the pure nodes in the tree. iv) What is the accuracy of the model on training data? v) What is the output of the decision tree if the test feature vector is [80, Overcast, Strong,85]		CO3	L4

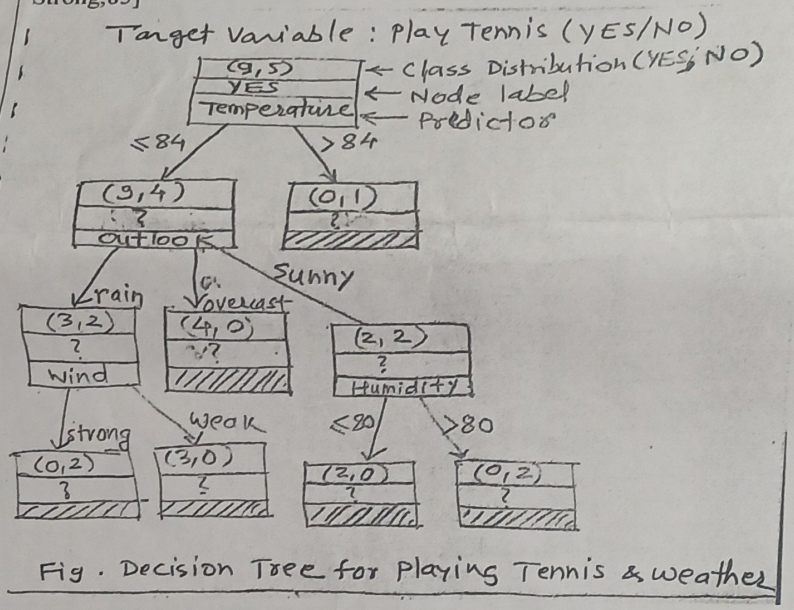
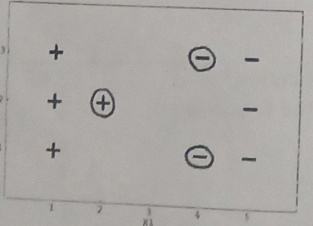


Fig. Decision Tree for playing Tennis & weather

	c)	Explain the following SVM terminologies using appropriate graph of Linear classification with two features x1 and x2. i) Support vectors ii) Maximum margin hyperplane iii) Margin iv) positive and negative hyperplanes Suppose you are using a Linear SVM classifier with binary classification problem. Now you have been given the following data in which some points are circled red that are representing support vectors. Then analyse the following vector space and answer the following question as True or False with Justification.	5	CO3	L4



i) If you remove the following any one red points from the data will the decision boundary change ?
ii) If you remove the non-red circled points from the data, will the decision boundary change?